

3070. Count Submatrices with Top-Left Element and Sum Less Than k

Solved

Medium

Topics

Companies

You are given a **0-indexed** integer matrix `grid` and an integer `k`.

Return the **number** of submatrices that contain the top-left element of the `grid`, and have a sum less than or equal to `k`.

Example 1:

7	6	3
6	6	1

7	6	3
6	6	1

7	6	3
6	6	1

7	6	3
6	6	1

Input: `grid = [[7,6,3],[6,6,1]]`, `k = 18`

Output: 4

Explanation: There are only 4 submatrices, shown in the image above, that contain the top-left element of `grid`, and have a sum less than or equal to 18.

Example 2:

7	2	9
1	5	0
2	6	6

7	2	9
1	5	0
2	6	6

7	2	9
1	5	0
2	6	6

7	2	9
1	5	0
2	6	6

7	2	9
1	5	0
2	6	6

7	2	9
1	5	0
2	6	6

Input: `grid = [[7,2,9],[1,5,0],[2,6,6]]`, `k = 20`

Output: 6

Explanation: There are only 6 submatrices, shown in the image above, that contain the top-left element of grid, and have a sum less than or equal to 20.

Constraints:

- `m == grid.length`
- `n == grid[i].length`
- `1 <= n, m <= 1000`
- `0 <= grid[i][j] <= 1000`
- `1 <= k <= 109`

Python:

#translated using AI

class Solution:

```
def countSubmatrices(self, grid, k):  
    n, m = len(grid), len(grid[0])
```

```

prefix = [0] * m
ans = 0

for i in range(n):
    rowSum = 0
    for j in range(m):
        rowSum += grid[i][j]
        prefix[j] += rowSum

    if prefix[j] <= k:
        ans += 1

return ans

```

JavaScript:

```

var countSubmatrices = function(grid, k) {
    let m = grid.length;
    let n = grid[0].length;

    let count = 0;

    for(let i = 0; i < m; i++){
        for(let j = 0; j < n; j++){
            if(i > 0) grid[i][j] += grid[i-1][j];
            if(j > 0) grid[i][j] += grid[i][j-1];
            if(i > 0 && j > 0) grid[i][j] -= grid[i-1][j-1];

            if(grid[i][j] <= k) count++;
        }
    }

    return count;
};

```

Java:

```

class Solution {
    public int countSubmatrices(int[][] grid, int k) {
        int n = grid.length, m = grid[0].length;

        int[] prefix = new int[m];
        int ans = 0;

        for(int i = 0; i < n; i++){
            int rowSum = 0;

```

```
        for(int j = 0; j < m; j++){
            rowSum += grid[i][j];
            prefix[j] += rowSum;

            if(prefix[j] <= k) ans++;
        }
    }
    return ans;
}
}
```